AegisAI: Designing Defense-in-Depth Safety for Conversational Text-to-Image Generation

Ngoc-Dai Tran[1,2]*, Thanh-Tuong Huynh[1,2]*, Hien Trinh[1,2], Minh-Triet Tran[1,2], and Trung-Nghia Le[1,2]**

[1] University of Science, Ho Chi Minh City, Vietnam
[2] Vietnam National University, Ho Chi Minh City, Vietnam
{22120045, 22120411, 23120254}@student.hcmus.edu.vn,
{tmtriet, ltnghia}@fit.hcmus.edu.vn

Abstract. Conversational text-to-image systems retain dialogue history, while many safety mechanisms evaluate individual prompts. We investigate this mismatch by evaluating seven representative text guardrails on the existing 10,000-conversation GuardChat benchmark, comparing final-prompt and full-dialogue inputs. Most guards show substantially higher attack success rates with conversational context, revealing a context-dependent safety gap that motivates multi-stage protection. We present AegisAI, model-agnostic middleware combining conversation-level input screening with independent post-generation visual moderation. It decouples the safety corridor from the generator and provides autonomous deployment with an acceptance gate that validates generated outputs before exposure. A within-subjects study with 20 users shows comparable overall usability, with costs in responsiveness and benign over-refusal. Participants rated policy-specific refusals as clear and reported greater willingness to adopt the guarded system. We derive five design implications for usable safety in conversational generative interfaces.

Keywords: Text-to-image generation, Guardrails, Multi-turn jailbreak, Generative AI safety

1 Introduction

Conversational T2I systems support iterative creation by conditioning each generation on dialogue history. This creates a safety challenge: harmful intent may emerge from otherwise innocuous turns rather than the current prompt alone. Many text-stage mechanisms, however, receive only the latest prompt even when the generator receives the full conversation, creating a context asymmetry. Recent work has highlighted multi-turn jailbreaks in language and image generation [47,33,41,38], but their impact across representative T2I guardrails remains less clear.

* These authors contributed equally to this work.
** Corresponding author.

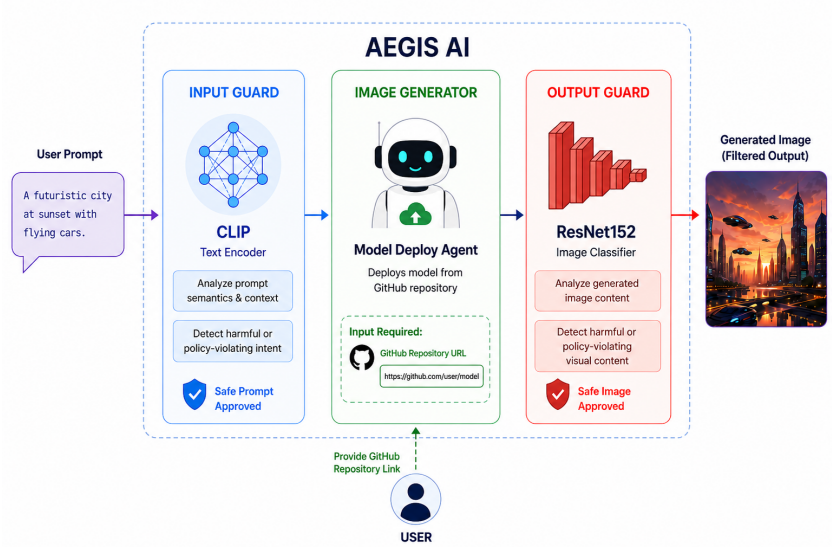


Fig. 1: AegisAI places a conversation-level Input Guard before a pluggable generator and a visual Output Guard after synthesis. Both guards are independent of the backend, including autonomously deployed generators.

We systematically evaluate seven representative text-stage guardrails using the existing 10,000-conversation GuardChat benchmark, comparing the final harmful prompt alone with the complete dialogue. This paired comparison isolates the effect of conversational context. Most guards show a substantial safety gap: single-turn ASR ranges from 5.50–21.30%, versus 10.50–80.45% for multi-turn inputs. SafeGuider increases from 5.50% to 58.21%, and BiLSTM from 14.22% to 80.45%, while Qwen3Guard-Gen-8B remains nearly stable. These findings identify available context as a critical safety variable and motivate explicit conversational coverage.

Motivated by this empirical finding, we propose **AegisAI**, a model-agnostic safety middleware with two independent boundaries: an Input Guard that evaluates the complete dialogue before synthesis and an Output Guard that independently screens the generated image. A decoupled generation contract makes the safety corridor independent of the underlying generator, allowing the same protection to cover heterogeneous backends, including newly deployed models. Fig. 1 summarizes the architecture.

Safety mechanisms can nevertheless introduce usability costs through interruptions, latency, and false positives, while transparent explanations may help users repair blocked requests. We therefore evaluate AegisAI as an interactive system. In a within-subjects study ($N = 20$), participants used the same backend with and without the safety corridor. Overall SUS was similar (79.4 vs. 82.1, $p = 0.18$), while responsiveness was lower ($p < 0.001$), benign over-refusal was 7.5%, and adoption intent was higher ($p = 0.014$). These results highlight the tension among protection, latency, and false positives.

Our contributions are as follows:

- We propose AegisAI, a model-agnostic defense-in-depth middleware that combines conversation-aware input screening with post-generation visual moderation.
- We evaluate the interaction consequences of guarded T2I generation through a within-subjects study, characterizing usability, responsiveness, workload, benign over-refusal, refusal transparency, and adoption intent.
- We synthesize five design implications for guarded generative interfaces, addressing latency, repairable refusals, visible defense in depth, over-refusal, and deployment safety.

2 Related Work

Safety Mechanisms in Text-to-Image Pipelines. Text-to-image safety mechanisms operate before, during, or after generation, using prompt filters and semantic classifiers for pre-generation screening [26], modifying latent representations or model parameters to suppress unsafe concepts [35,10], or inspecting synthesized images through visual moderation [23,36]. Reusable guardrail models and orchestration frameworks further decouple moderation from application logic [30,17,44,46], improving portability across systems. However, these existing approaches remain tied either to a particular generator or to a single stage of inference, therefore motivating AegisAI to place independent text and visual boundaries outside the generator, allowing both guards to evolve separately while protecting heterogeneous backends through the same safety corridor.

Multi-Turn, Memory-Exploiting Attacks. Single-turn jailbreaks struggle with an inherent trade-off between semantic retention and detection evasion [38], an obstacle multi-turn strategies dissolve by distributing obfuscation across conversational turns. Recent attacks against language models demonstrate progressive escalation [33,41], conversational framing [1,24], and substantial human red-teaming gains over automated single-turn baselines [22], while visual attacks decompose illicit targets into iterative editing steps [38], exploiting conversational memory itself as a primary vulnerability [47]. However, standard T2I safety suites remain exclusively single-turn [35,27,20,45] and multi-turn benchmarks focus almost entirely on text-to-text models [13,22,43]. Rather than introducing another benchmark, our work uses the existing GuardChat benchmark as a common evaluation substrate for a systematic comparison of seven representative text-stage guardrails. We compare the same conversational attacks under final-prompt-only and full-dialogue conditions, and use the resulting safety gap to derive requirements for AegisAI.

Human Side of Moderation. While content moderation literature shows that transparent explanations foster behavioral compliance whereas opaque enforcement provokes contestation [12,18,31,37], generative interfaces relocate this dynamic directly into the user's creative loop. Existing exaggerated-safety benchmarks quantify over-refusal frequencies across language [7,32] and image models [6] without evaluating perceptual downstream impacts on user trust [19,21,25],

Table 1: Our evaluation of seven representative text-stage guardrails on the existing 10,000-conversation GuardChat benchmark. We compare access to only the final prompt (single-turn) with access to the complete chronological conversation (multi-turn).

Guardrail	**Single-turn ASR (%) ↓**	**Multi-turn ASR (%) ↓**	**Macro-F1 (multi-turn) ↑**
BiLSTM	14.22	80.45	0.312
BERT [8]	9.85	72.30	0.422
SafeGuider [26]	**5.50**	58.21	0.487
Qwen2.5-7B [28]	7.90	60.33	0.542
Llama-Guard-3-8B [17]	6.50	52.15	**0.579**
ShieldGemma-2B (τ=0.7) [44]	21.30	39.70	0.273
Qwen3Guard-Gen-8B [46]	11.00	**10.50**	0.223

which human–AI guidelines suggest mitigating by maintaining system limits legible at runtime [2]. Complementing developer-oriented responsible-AI toolkits [11,39,40], AegisAI enforces an unbypassable platform-level corridor, measuring objective over-refusal alongside subjective user responses against a matched unguarded baseline to evaluate the holistic interaction costs of real-time intervention.

3 Preliminary Analysis

3.1 Findings

We use the existing GuardChat dataset [3], comprising 10,000 prompt–conversation pairs with 6–10 turns and six harm categories: `SEXUAL`, `VIOLENCE`, `HARASSMENT`, `SHOCKING`, `ILLEGAL`, and `SELF_HARM`. Each conversation was verified by the benchmark creators to induce a policy-violating image. We hold each attack conversation fixed and evaluate seven representative text-stage guards under two input conditions: (1) the final prompt alone (single-turn) and (2) the complete chronological dialogue (multi-turn).

Table 1 summarizes our evaluation. Six of the seven evaluated guardrails show a substantial increase in Attack Success Rate (ASR) when the complete conversation is provided, demonstrating that conversational context can fundamentally change safety behavior. SafeGuider [26] , for example, increases from 5.50% ASR with the final prompt alone to 58.21% with the full conversation, while BiLSTM increases from 14.22% to 80.45%. BERT [8], Qwen2.5-7B [28], Llama-Guard-3-8B [17], and ShieldGemma-2B [44] also exhibit substantial increases. Qwen3Guard-Gen-8B [46] is an important exception: its ASR changes only from 11.00% to 10.50%, consistent with its explicit multi-turn moderation capability. Thus, the findings should not be interpreted as evidence that text guardrails are inherently inadequate; rather, they show that *context coverage* is an important architectural property of conversational moderation.

[3] `https://huggingface.co/datasets/multimedia-synergy-lab/GuardChat`

The results also reveal a second design requirement. Even a context-aware text guard operates before image synthesis and therefore cannot directly inspect the visual artifact produced by the generator. Residual failures may arise from imperfect text classification or generator-specific behavior. We therefore treat the input guard as the first boundary rather than the complete safety solution. Finally, the Macro-F1 results highlight the distinction between detecting a violation and assigning a category suitable for user-facing intervention: Qwen3Guard has low multi-turn ASR but also low Macro-F1 because its taxonomy does not directly cover all six categories.

3.2 Design Goals

The preliminary evaluation leads to five design goals for AegisAI.

G1: Match the safety context to the generation context. The first guard should process the complete dialogue available to the generator rather than only the latest prompt.

G2: Establish an independent visual boundary. A second guard should inspect the generated image so that failures at the text stage do not directly expose unsafe visual outputs.

G3: Decouple safety from generation. The safety corridor should operate independently of the underlying T2I architecture, allowing heterogeneous generators to be protected without retraining or modifying their internal models.

G4: Make refusals repairable. Blocked requests should receive structured, policy-specific explanations that help users understand and revise the request.

G5: Preserve the safety boundary during deployment. Newly mounted generation backends should enter the same guarded execution path rather than bypassing the established safety mechanisms.

4 Proposed AegisAI System

4.1 Architecture

AegisAI is middleware between the user and a T2I backend (Fig. 1). Its architecture follows two requirements from our preliminary evaluation: safety mechanisms should access the generator's conversational context, and text screening cannot inspect the final visual artifact. The Input Guard receives the prompt and dialogue history and returns a safety verdict and category before synthesis. The Output Guard independently classifies the generated image and releases it only when labeled `SAFE`. A decoupled generation engine provides a common interface to remote APIs, self-hosted models, and repositories.

4.2 Input Guard: Screening the Conversation

The Input Guard addresses the context asymmetry identified in our preliminary study. It concatenates the full dialogue and passes it through a CLIP text

Table 2: Visual moderation backbones on the held-out safety test set.

Model	Accuracy	Precision	Recall	F1
ViT-B/16 [9]	0.8347	0.8128	0.6789	0.7398
MobileNetV2 [34]	0.9009	0.9030	0.8992	0.8993
ResNet-152 [16]	**0.9307**	**0.8792**	**0.9254**	**0.9017**

encoder. The final `[EOS]` representation feeds a three-layer MLP fine-tuned on GuardChat to predict safe/unsafe status and one of six harm categories. Safe requests proceed; blocked requests receive a structured, category-labelled refusal. The default SafeGuider implementation is lightweight because it reuses the CLIP representation [26,29]. This design does not claim to eliminate jailbreaks; it ensures that the input boundary accesses the generator's contextual information, while the independent Output Guard provides a second boundary.

4.3 Output Guard: Screening the Pixels

The Output Guard provides an independent visual boundary. We compare three backbones on the same visual safety dataset (Tab. 2). ResNet-152 has the highest recall (0.9254) and is used as the default; MobileNetV2 remains suitable for constrained deployments. At runtime, images are classified into `SAFE` plus the six harm categories. Only `SAFE` outputs are returned; rejected image data are discarded.

4.4 Decoupled Generation and Self-Service Deployment

All backends implement one generation contract for the prompt, optional parameters, and a decodable image. A router translates this contract to provider-specific APIs, isolating moderation from the generation model and enabling backend switching without changing safety logic.

Autonomous deployment. Users can provide an image-generation repository. The deployment pipeline performs intake, triage, provisioning, wrapper adaptation, launch, and a fixed acceptance test; failures enter a bounded repair-or-rotate loop (Fig. 2). The agent may edit only the serving wrapper, while the acceptance test remains outside its edit surface. Execution is isolated, dependencies are pinned, secrets remain external, and public exposure requires human review. A passing gate certifies generator functionality, not source-code benignity.

4.5 Interface

AegisAI provides a chat interface with backend selection, generation, inline refusals, and a deployment dashboard (Fig. 3). For compliant requests the guards are largely invisible; for blocked requests the interface reports the violated policy.

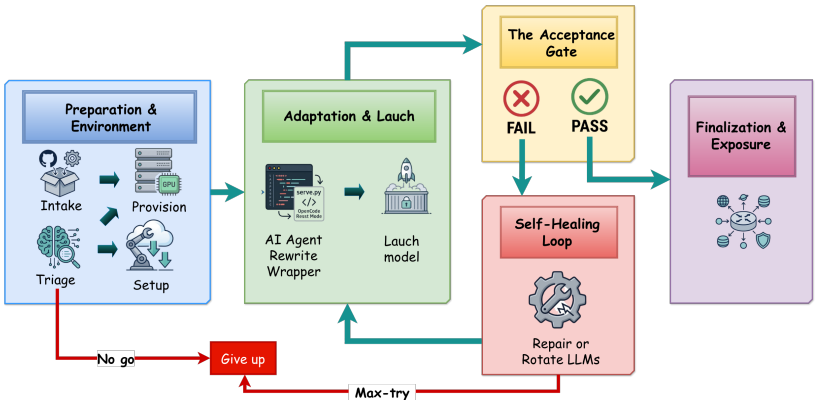


Fig. 2: Autonomous deployment: intake and triage precede adaptation and launch; a fixed acceptance gate validates the generated output before exposure, with bounded self-healing on failure.

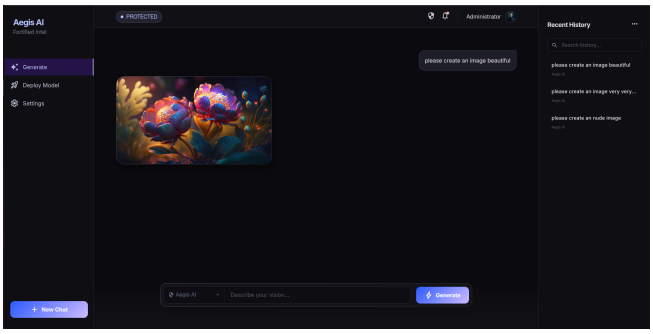


Fig. 3: Guarded chat interface with backend selection, generated outputs, and inline policy refusals.

5 User Study

The preliminary evaluation motivates conversation-level protection, while the system design establishes the resulting defense architecture. We next examine how this safety corridor affects users of a conversational T2I system. We compare AegisAI with the identical backend without either guard, isolating middleware effects from differences in image-generation capability.

5.1 Research Questions

While Sec. 3 evaluates attack mitigation, our user study examines perceptual and interaction consequences across four questions:

RQ1: How does safety middleware alter users' perceived protection compared to an unguarded system?

RQ2: What operational costs does real-time moderation impose on usability, responsiveness, and subjective workload?
RQ3: How frequently does the system trigger benign over-refusals, and how transparent and acceptable are its explanatory interventions?
RQ4: To what extent does the unified serving interface facilitate usable self-service deployment for external generators?

5.2 Settings

Participants. We recruited 20 participants (10 male, 10 female), aged 18–24, balanced between technical and non-technical backgrounds. All had prior T2I experience and none had used AegisAI. They completed benign and adversarially framed single- and multi-turn tasks.
Design and Procedure. We used a within-subjects design with two conditions: **guarded**, with both guards enabled, and **unguarded**, using the same backend with both guards disabled. Thus, the comparison isolates the moderation corridor rather than image-generation differences. Condition order was also counterbalanced, reducing the systematic order effects.

Before the experimental tasks, participants provided informed consent, completed a demographic questionnaire, and received a brief interface practice task. They then completed a scripted task set with three prompt categories: benign, overt single-turn harmful, and multi-turn deceptive conversations. Both conditions used the same task set. In the unguarded condition, generated images were returned directly without either guard; consequently, participants could see outputs that the guarded pipeline would have blocked. Exposure was confined to the scripted session, and participants could stop at any time. A separate task asked participants to deploy a previously unseen image-generation model from a repository link.
Measures and Analysis. Throughout each session, screen activity was recorded to support retrospective stimulated recall. Following the completion of tasks under each condition, quantitative feedback was collected via the SUS [5,3], NASA-TLX [15,14] and a custom 7-point Likert scale evaluating protection, refusal clarity/acceptability, responsiveness, and adoption. For the deployment task specifically, we also recorded task completion time and objective success rates, followed by a separate SUS. Participants then reviewed their screen recordings in a retrospective stimulated recall session to reflect on trust, friction, and decision-making processes. For data analysis, paired comparisons were evaluated using the Wilcoxon signed-rank test [42], while qualitative responses were analyzed thematically [4].

5.3 Results

Usability and Interaction Experience (RQ2). Overall SUS did not differ significantly (Fig. 4): 79.4 ($SD = 8.6$) for AegisAI versus 82.1 ($SD = 7.9$) unguarded ($p = 0.18$). Ease of use and image satisfaction were also similar. Responsiveness was lower with guarding (5.30 vs. 6.00, $p < 0.001$), while adoption intent was higher (6.00 vs. 5.70, $p = 0.014$).

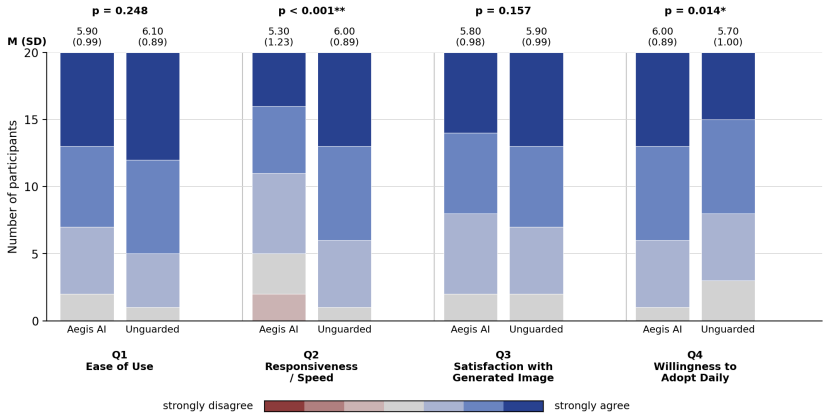


Fig. 4: Interaction questionnaire on a 7-point Likert scale ($N = 20$). Ease of use and image satisfaction do not differ; responsiveness is lower and adoption intent is higher in the guarded condition.

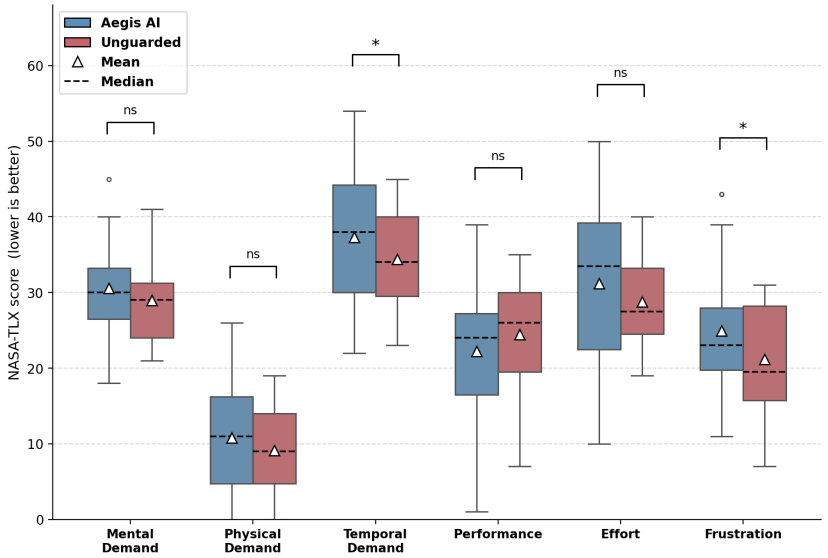


Fig. 5: NASA-TLX across six subscales. Only temporal demand and frustration differ significantly.

Task Load (RQ2). Mental demand, physical demand, performance, and effort did not differ significantly (Fig. 5). Temporal demand and frustration were slightly higher under guarding ($p < 0.05$), consistent with added latency and benign over-refusal.

Table 3: Over-refusal and refusal-transparency ratings.

Metric	AegisAI	Unguarded
Over-refusal rate on benign prompts (%)	7.5	0.8
Refusal clarity (7-pt Likert)	6.1	—
Refusal acceptability (7-pt Likert)	5.6	—

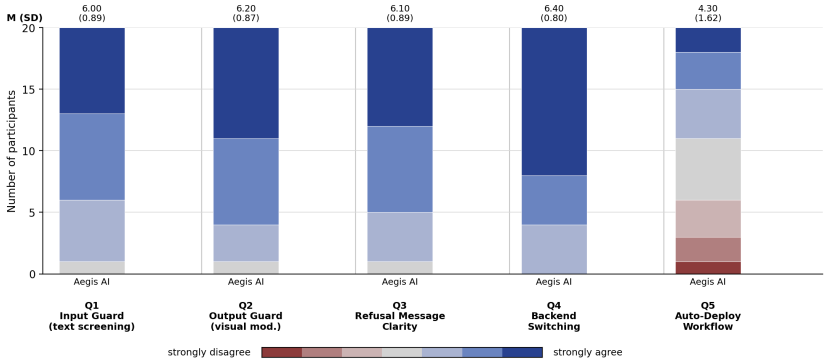


Fig. 6: Component ratings on a 7-point Likert scale. Autonomous deployment receives the lowest rating.

Over-Refusal and Refusal Transparency (RQ3). Tab. 3 shows that the guarded pipeline blocked 7.5% of benign requests, versus 0.8% in the unguarded condition. Participants rated category-labelled refusals as clear ($M = 6.1$) and acceptable ($M = 5.6$). The remaining friction therefore came primarily from false positives rather than from the presence of moderation itself.

Component Ratings and Deployment (RQ4). Fig. 6 shows that conversation-level input screening ($M = 6.00$), visual moderation ($M = 6.20$), refusal clarity ($M = 6.10$), and backend switching ($M = 6.40$) were rated highly. Autonomous deployment was lower ($M = 4.30$, $SD = 1.62$). Only 55% of deployment attempts reached `Success`. The reported 14.5-min median time to a live endpoint is calculated over the successful deployment observations only; failed attempts are unsuccessful/censored observations for this endpoint-time measure and therefore do not contribute a completion time (Tab. 4). The median number of attempts before success was 2, and deployment SUS was 68.5. Failures were associated mainly with repository quality (missing documentation, dependencies, or inference entry points) and infrastructure limits.

Qualitative Findings (RQ1, RQ3). Three themes explain the quantitative results. First, participants valued refusals that named the violated policy because they could revise the prompt. Second, seeing the Output Guard intercept an unsafe image increased perceived protection. Third, frustration centered on benign over-blocking rather than on moderation as such. Participants also val-

Table 4: Autonomous deployment results ($N = 20$). The 14.5-min median refers to attempts reaching a live endpoint.

Metric	Value
Task-success rate (%)	55.0
Median time to live endpoint (min)	14.5
Median deployment attempts before success	2
Deployment dashboard SUS score	68.5

ued self-service deployment but reported lower confidence after failed or slow deployments.

6 Discussion

Conversational Context Is a Safety Variable. Our preliminary evaluation shows that safety performance depends on the context presented to the guardrail. The same attack conversation can produce different outcomes when the guard receives only the final prompt versus the complete dialogue. Across seven guards, Qwen3Guard-Gen-8B remains stable under both conditions. Thus, contextual coverage is an important architectural property rather than evidence that text guardrails are inherently inadequate.

Safety Can Increase Adoption While Increasing Friction. The user study reveals a tension: participants perceived the guarded system as less responsive and reported higher temporal demand, yet adoption intent was higher. Participants valued visible Output Guard intervention and category-specific refusals for repairing blocked requests. This suggests that safety interventions benefit from being visible, understandable, and actionable rather than opaque.

Moderation Cost Is Concentrated in Time and Over-Blocking. The guarded corridor did not significantly change overall SUS, ease of use, image satisfaction, mental demand, effort, or perceived performance. Its measurable costs were responsiveness, temporal demand, and false positives, highlighting latency and over-refusal as primary optimization targets.

Refusal Is Part of the Interaction. Clarity (6.1) exceeded acceptability (5.6), indicating that understanding a refusal does not require agreeing with it. Category-labelled responses provide an actionable next step, whereas a generic block provides little guidance, making category attribution a user-interface concern in addition to a classification metric.

Defense in Depth and Extensibility. Participants associated the Output Guard with protection because they could observe a second boundary catch content that passed the text stage. Backend switching was rated highly, whereas autonomous deployment was lower; its current success rate and latency limit self-service onboarding.

Design Implications. These findings suggest five design principles: overlap post-generation inspection with synthesis to reduce perceived latency; provide category-specific explanations that make refusals repairable; report benign over-refusal alongside attack mitigation; make visual-boundary interventions legible; and enforce independent deployment validation so onboarding agents cannot certify their own integration.

Limitations. The study uses 20 young participants with prior T2I experience, limiting generalisability. The identical-backend baseline isolates moderation cost but does not compare alternative moderation designs. Per-item tests were not corrected for multiple comparisons, and stochastic generation and moderation add run-to-run variance.

The same task set was used in both conditions; although order was counterbalanced, item-level carryover remains possible. No per-task SEQ was collected, so friction cannot be localized to task categories. Exact category prompt counts are not reported. The unguarded condition exposed participants to unfiltered outputs, limiting ecological validity. Finally, the 55% deployment success rate describes the current implementation, while the 14.5-min endpoint time is based only on successful deployments.

7 Conclusion

We investigated safety in conversational text-to-image generation from technical and interaction perspectives. On the existing 10,000-conversation benchmark, we evaluated seven text-stage guardrails under single-turn and full-conversation conditions. Most showed substantially higher attack success rates with conversational context, revealing a context-dependent safety gap. This result motivated AegisAI, model-agnostic defense-in-depth middleware combining conversation-level input screening with independent visual moderation.

Our user study further shows that safety mechanisms become part of the interaction. AegisAI preserved overall usability but introduced responsiveness costs and benign over-refusal, while policy-specific refusals were perceived as clear and willingness to adopt increased. Effective safety therefore requires both aligned safety boundaries and carefully designed user-facing interventions.

Ethics and Broader Impact Participants were informed before the session that the study included harmful and adversarial prompts and could stop at any time. In the unguarded condition, generated images were returned without filtering, so participants could encounter outputs that the guarded pipeline would have blocked; exposure was limited to the scripted tasks. Our benchmark analysis reports aggregate guard behaviour without releasing attack prompts or exploitation recipes. The main practical risk observed in the user study is over-refusal: a 7.5% benign-block rate can exclude legitimate requests and should therefore be reported alongside safety metrics.

Acknowledgments. This research is funded by Vietnam National University - Ho Chi Minh City (VNU-HCM) under Grant Number B2026-18-17.

References

1. Alobaid, A., Roca, M.J., Castillo, C., Vendrell, J.: The echo chamber multi-turn llm jailbreak. arXiv preprint arXiv:2601.05742 (2026)
2. Amershi, S., Weld, D., Vorvoreanu, M., Fourney, A., Nushi, B., Collisson, P., Suh, J., Iqbal, S., Bennett, P.N., Inkpen, K., Teevan, J., Kikin-Gil, R., Horvitz, E.: Guidelines for human-ai interaction. In: Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems (CHI). pp. 1–13 (2019). https://doi.org/10.1145/3290605.3300233
3. Bangor, A., Kortum, P.T., Miller, J.T.: An empirical evaluation of the system usability scale. International Journal of Human-Computer Interaction **24**(6), 574–594 (2008)
4. Braun, V., Clarke, V.: Using thematic analysis in psychology. Qualitative Research in Psychology **3**(2), 77–101 (2006)
5. Brooke, J.: SUS: A "quick and dirty" usability scale. In: Usability Evaluation in Industry, pp. 189–194. Taylor & Francis, London, UK (1996)
6. Cheng, Z., Huang, Y., Xu, H., Sojoudi, S., Zhao, X., Song, D., Mei, S.: OVERT: A benchmark for over-refusal evaluation on text-to-image models. Advances in Neural Information Processing Systems **38** (2026)
7. Cui, J., Chiang, W.L., Stoica, I., Hsieh, C.J.: OR-Bench: An over-refusal benchmark for large language models. In: Proceedings of the 42nd International Conference on Machine Learning (ICML). Proceedings of Machine Learning Research, vol. 267, pp. 11515–11542. PMLR (2025)
8. Devlin, J., Chang, M.W., Lee, K., Toutanova, K.: Bert: Pre-training of deep bidirectional transformers for language understanding. In: Proceedings of the 2019 conference of the North American chapter of the association for computational linguistics: human language technologies, volume 1 (long and short papers). pp. 4171–4186 (2019)
9. Dosovitskiy, A., Beyer, L., Kolesnikov, A., Weissenborn, D., Zhai, X., Unterthiner, T., Dehghani, M., Minderer, M., Heigold, G., Gelly, S., Uszkoreit, J., Houlsby, N.: An image is worth 16x16 words: Transformers for image recognition at scale. In: International Conference on Learning Representations (ICLR) (2021)
10. Gandikota, R., Materzynska, J., Fiotto-Kaufman, J., Bau, D.: Erasing concepts from diffusion models. In: Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV). pp. 2426–2436 (2023)
11. Ganguli, D., Lovitt, L., Kernion, J., Askell, A., Bai, Y., Kadavath, S., Mann, B., Perez, E., Schiefer, N., Ndousse, K., et al.: Red teaming language models to reduce harms: Methods, scaling behaviors, and lessons learned. arXiv preprint arXiv:2209.07858 (2022)
12. Gillespie, T.: Custodians of the Internet: Platforms, Content Moderation, and the Hidden Decisions That Shape Social Media. Yale University Press, New Haven, CT (2018)
13. Ha, J., Kim, H., Yu, S., Park, H., Yousefpour, A., Park, Y., Kim, S.: M2S: Multi-turn to single-turn jailbreak in red teaming for LLMs. In: Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (ACL). vol. 1, pp. 16489–16507 (2025)
14. Hart, S.G.: NASA-Task Load Index (NASA-TLX): 20 years later. In: Proceedings of the Human Factors and Ergonomics Society Annual Meeting. vol. 50, pp. 904–908. SAGE (2006)

15. Hart, S.G., Staveland, L.E.: Development of NASA-TLX (task load index): Results of empirical and theoretical research. In: Human Mental Workload, Advances in Psychology, vol. 52, pp. 139–183. Elsevier (1988)
16. He, K., Zhang, X., Ren, S., Sun, J.: Deep residual learning for image recognition. In: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR). pp. 770–778 (2016)
17. Inan, H., Upasani, K., Chi, J., Rungta, R., Iyer, K., Mao, Y., Tontchev, M., Hu, Q., Fuller, B., Testuggine, D., Khabsa, M.: Llama guard: LLM-based input-output safeguard for human-ai conversations. arXiv preprint arXiv:2312.06674 (2023)
18. Jhaver, S., Bruckman, A., Gilbert, E.: Does transparency in moderation really matter? user behavior after content removal explanations on reddit. Proceedings of the ACM on Human-Computer Interaction **3**(CSCW) (2019). `https://doi.org/10.1145/3359252`
19. Jian, J.Y., Bisantz, A.M., Drury, C.G.: Foundations for an empirically determined scale of trust in automated systems. International Journal of Cognitive Ergonomics **4**(1), 53–71 (2000)
20. Jin, X., Weng, Z., Guo, H., Yin, C., Cheng, S., Shen, G., Zhang, X.: JailbreakDiffBench: A comprehensive benchmark for jailbreaking diffusion models. In: Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV). pp. 16461–16471 (2025)
21. Lee, J.D., See, K.A.: Trust in automation: Designing for appropriate reliance. Human Factors **46**(1), 50–80 (2004)
22. Li, N., Han, Z., Steneker, I., Primack, W., Goodside, R., Zhang, H., Wang, Z., Menghini, C., Yue, S.: LLM defenses are not robust to multi-turn human jailbreaks yet. arXiv preprint arXiv:2408.15221 (2024)
23. Markov, T., Zhang, C., Agarwal, S., Eloundou, T., Lee, T., Adler, S., Jiang, A., Weng, L.: A holistic approach to undesired content detection in the real world. In: Proceedings of the AAAI Conference on Artificial Intelligence. vol. 37, pp. 15009–15018 (2023)
24. Panda, S., Rai, A.: Say it differently: Linguistic styles as jailbreak vectors. arXiv preprint arXiv:2511.10519 (2025)
25. Parasuraman, R., Riley, V.: Humans and automation: Use, misuse, disuse, abuse. Human Factors **39**(2), 230–253 (1997)
26. Qi, P., Tang, K., Zhou, W., Zhang, W., Yu, N., Zhang, T., Guo, Q., Zhang, J.: SafeGuider: Robust and practical content safety control for text-to-image models. In: Proceedings of the 2025 ACM SIGSAC Conference on Computer and Communications Security (CCS). pp. 2818–2832 (2025)
27. Quaye, J., Parrish, A., Inel, O., Rastogi, C., Kirk, H.R., Kahng, M., Van Liemt, E., Bartolo, M., Tsang, J., White, J., et al.: Adversarial nibbler: An open red-teaming method for identifying diverse harms in text-to-image generation. In: Proceedings of the 2024 ACM Conference on Fairness, Accountability, and Transparency (FAccT). pp. 388–406 (2024)
28. Qwen, :, Yang, A., Yang, B., Zhang, B., Hui, B., Zheng, B., Yu, B., Li, C., Liu, D., Huang, F., Wei, H., Lin, H., Yang, J., Tu, J., Zhang, J., Yang, J., Yang, J., Zhou, J., Lin, J., Dang, K., Lu, K., Bao, K., Yang, K., Yu, L., Li, M., Xue, M., Zhang, P., Zhu, Q., Men, R., Lin, R., Li, T., Tang, T., Xia, T., Ren, X., Ren, X., Fan, Y., Su, Y., Zhang, Y., Wan, Y., Liu, Y., Cui, Z., Zhang, Z., Qiu, Z.: Qwen2.5 technical report. arXiv preprint arXiv:2412.15115 (2025)
29. Radford, A., Kim, J.W., Hallacy, C., Ramesh, A., Goh, G., Agarwal, S., Sastry, G., Askell, A., Mishkin, P., Clark, J., Krueger, G., Sutskever, I.: Learning transfer-

able visual models from natural language supervision. In: Proceedings of the 38th International Conference on Machine Learning (ICML). pp. 8748–8763 (2021)
30. Rebedea, T., Dinu, R., Sreedhar, M.N., Parisien, C., Cohen, J.: NeMo guardrails: A toolkit for controllable and safe LLM applications with programmable rails. In: Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing: System Demonstrations. pp. 431–445 (2023)
31. Roberts, S.T.: Behind the Screen: Content Moderation in the Shadows of Social Media. Yale University Press, New Haven, CT (2019)
32. Röttger, P., Kirk, H., Vidgen, B., Attanasio, G., Bianchi, F., Hovy, D.: XSTest: A test suite for identifying exaggerated safety behaviours in large language models. In: Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (NAACL-HLT). pp. 5377–5400 (2024). https://doi.org/10.18653/v1/2024.naacl-long.301
33. Russinovich, M., Salem, A., Eldan, R.: Great, now write an article about that: The crescendo multi-turn llm jailbreak attack. In: 34th USENIX Security Symposium (USENIX Security 25). pp. 2421–2440 (2025)
34. Sandler, M., Howard, A., Zhu, M., Zhmoginov, A., Chen, L.C.: MobileNetV2: Inverted residuals and linear bottlenecks. In: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR). pp. 4510–4520 (2018)
35. Schramowski, P., Brack, M., Deiseroth, B., Kersting, K.: Safe latent diffusion: Mitigating inappropriate degeneration in diffusion models. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR). pp. 22522–22531 (2023)
36. Schramowski, P., Tauchmann, C., Kersting, K.: Can machines help us answering question 16 in datasheets, and in turn reflecting on inappropriate content? In: Proceedings of the 2022 ACM Conference on Fairness, Accountability, and Transparency (FAccT). pp. 1350–1361 (2022). https://doi.org/10.1145/3531146.3533192
37. Vaccaro, K., Sandvig, C., Karahalios, K.: "at the end of the day facebook does what it wants": How users experience contesting algorithmic content moderation. Proceedings of the ACM on Human-Computer Interaction **4**(CSCW2) (2020). https://doi.org/10.1145/3415238
38. Wang, W., Gao, K., Yuan, Y., Huang, J.t., Liu, Q., Wang, S., Jiao, W., Tu, Z.: Chain-of-jailbreak attack for image generation models via step by step editing. In: Findings of the Association for Computational Linguistics: ACL 2025. pp. 10940–10957 (2025)
39. Wang, Z.J., Kulkarni, C., Wilcox, L., Terry, M., Madaio, M.: Farsight: Fostering responsible ai awareness during ai application prototyping. In: Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems (CHI) (2024). https://doi.org/10.1145/3613904.3642335
40. Weidinger, L., Uesato, J., Rauh, M., Griffin, C., Huang, P.S., Mellor, J., Glaese, A., Cheng, M., Balle, B., Kasirzadeh, A., et al.: Taxonomy of risks posed by language models. In: Proceedings of the 2022 ACM Conference on Fairness, Accountability, and Transparency (FAccT). pp. 214–229 (2022). https://doi.org/10.1145/3531146.3533088
41. Weng, Z., Jin, X., Jia, J., Zhang, X.: Foot-in-the-door: A multi-turn jailbreak for llms. In: Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing (EMNLP). pp. 1939–1950 (2025)
42. Wilcoxon, F.: Individual comparisons by ranking methods. In: Breakthroughs in Statistics: Methodology and Distribution, pp. 196–202. Springer (1992)

43. Yang, X., Xiao, L., Li, S., Ladhak, F., Yun, H., Petzold, L.R., Xu, Y., Wang, W.Y.: Many-turn jailbreaking. arXiv preprint arXiv:2508.06755 (2025)
44. Zeng, W., Liu, Y., Mullins, R., Peran, L., Fernandez, J., Harkous, H., Narasimhan, K., Proud, D., Kumar, P., Radharapu, B., Sturman, O., Wahltinez, O.: Shield-Gemma: Generative ai content moderation based on gemma. arXiv preprint arXiv:2407.21772 (2024)
45. Zhang, C., Zhang, T., Wang, L., Chen, R., Li, W., Liu, A.: T2I-RiskyPrompt: A benchmark for safety evaluation, attack, and defense on text-to-image models. In: Proceedings of the AAAI Conference on Artificial Intelligence. vol. 40, pp. 36039–36047 (2026)
46. Zhao, H., Yuan, C., Huang, F., Hu, X., Zhang, Y., Yang, A., Yu, B., Liu, D., Zhou, J., Lin, J.: Qwen3Guard technical report. arXiv preprint arXiv:2510.14276 (2025)
47. Zhao, S., Liu, J., Li, Y., Hu, R., Jia, X., Fan, W., Wu, X., Li, X., Zhang, J., Dong, W., et al.: When memory becomes a vulnerability: Towards multi-turn jailbreak attacks against text-to-image generation systems. arXiv preprint arXiv:2504.20376 (2025)